

Comparing Simplified Representations of a Discrete Ising System

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Abstract—For stochastic, high-dimensional nonlinear systems, the word *linearization* can refer to several distinct simplification strategies rather than only a Jacobian approximation at a point. This paper uses a discrete-time Ising system with Glauber-style dynamics as a common testbed for comparing five models: one reference model and four lower-complexity surrogates. The surrogates span three broad representation classes—coarse observable models, probability-based models, and operator-level models. Our goal is not to identify a universally best simplification, but to clarify which features of the reference dynamics each representation preserves.

We compare the models in terms of transient magnetization behavior, steady-state bias, variance structure, interpretability, and computational tractability under shared parameter settings and common initial conditions. The main empirical finding is that visually similar long-time magnetization can arise from models with substantially different transient rates and variability. In the representative cases shown here, the affine probability operator most closely tracks the reference transient shape, the local-neighborhood and full-state probabilistic models exhibit slower convergence, and the full-state model exhibits the largest variance in several parameter regimes. These differences are quantified through trajectory RMSE, steady-state bias, variance mismatch, convergence-time difference, and computational cost.

The contribution is therefore methodological. By placing several simplifications of the same underlying Ising dynamics into a common experimental framework, the paper makes explicit the tradeoffs among fidelity, complexity, and interpretability, and shows where cross-model agreement is reliable and where it begins to fail.¹

I. INTRODUCTION

Local Jacobian linearization is the default simplification tool in low-dimensional deterministic systems. For stochastic and high-dimensional systems, however, there is no single canonical notion of simplification. One may preserve the full configuration-level probability law, track only marginal probabilities, or compress the system to a small set of coarse observables. These modeling choices are not equivalent: they preserve different information, support different analyses, and introduce different failure modes.

This paper studies that point in a concrete setting. We use a discrete Ising system with stochastic spin updates as a common reference process and compare several simplified models derived from, or motivated by, the same underlying

dynamics. The Ising model is a useful testbed because it is rich enough to exhibit collective behavior, sensitivity to coupling, and substantial differences between microscopic and macroscopic descriptions, yet small enough to simulate repeatedly over a controlled parameter grid [1], [2], [4].

The central question is simple: *how do different simplified representations of the same Ising dynamics differ in their ability to preserve transient behavior, steady-state behavior, and variability?* We are not primarily interested in proving that one model is universally superior. Rather, we want to expose how representation choice changes what the model can faithfully reproduce.

The paper makes four contributions. First, it frames a five-model comparison study organized into three representation classes. Second, it introduces an explicit taxonomy that separates coarse observable, probability-based, and operator-level approximations. Third, it compares the models under a shared protocol using both visual and quantitative metrics. Fourth, it highlights calibration and tractability as part of the modeling problem.

II. RELATED WORK AND BACKGROUND

A. The Ising setting

The Ising model describes a collection of binary spins coupled through a graph. Each spin takes values in $\{-1, +1\}$, neighboring spins influence one another through couplings, an external field biases the system globally, and temperature controls stochasticity [1], [2]. In ferromagnetic regimes, positive coupling encourages alignment and yields nonzero magnetization. In higher-temperature or weak-coupling regimes, local correlations weaken and the system behaves more like noise. Near critical regimes, coarse summaries can become informative yet incomplete because behavior emerges across multiple scales [3]–[5].

B. Why multiple simplifications matter

For high-dimensional stochastic systems, no single simplification captures every aspect of the original process. Mean-field models compress the system to a few global observables and gain interpretability at the cost of local structure. Probability-based models preserve more detail but typically require stronger independence assumptions or careful calibration. Operator-level models can be more faithful for small

¹Our GitHub repository [10], and the text of this report was produced with the help of OpenAI’s ChatGPT 5.3 Extended Reasoning model and Codex.

systems but often scale poorly because the state dimension grows combinatorially. Other approximations may be desirable for obtaining gradient information in the dynamics.

This tension appears well beyond statistical physics. In control, system identification, and reduced-order modeling, one routinely trades fidelity for tractability and interpretability [6]. In Koopman-style and observable-based approaches, one similarly changes representation in order to obtain simpler evolution laws for selected observables [7], [8]. The present paper is not a new reduction theory for Ising systems; instead, it is a controlled comparison of several familiar simplification strategies within one shared experiment.

C. What is standard and what is new here

The reference Ising dynamics and the general ideas of mean-field approximation, probabilistic state evolution, and coarse-grained modeling are standard [1], [2], [4], [9]. The novelty here is the side-by-side comparison of five concrete representations, the explicit organization of those models into representation classes, and the use of shared metrics to compare transient agreement, steady-state agreement, variance structure, and computational cost.

III. PROBLEM STATEMENT

We compare one reference model and four simplified models constructed from the same discrete Ising system. The reference observable is the magnetization trajectory

$$m_k := \frac{1}{N} \sum_{i=1}^N s_i(k), \quad (1)$$

where $s_i(k) \in \{-1, +1\}$ is the spin at site i and time k .

The paper is organized around the following claims.

- 1) Different simplifications can agree on coarse magnetization trends while disagreeing on transient rate and variance.
- 2) Increasing interpretability or tractability typically reduces the ability to preserve detailed structure.
- 3) Cross-model calibration is informative not only when it succeeds, but also where and how it fails.

Accordingly, each surrogate model is judged against the reference system by its ability to reproduce: (i) transient magnetization trajectories, (ii) steady-state magnetization, (iii) variability across repeated runs, and (iv) computational cost.

IV. REFERENCE DYNAMICS

Let $G = (V, E)$ be a finite interaction graph with $|V| = N$ sites. At time k , the spin configuration is

$$s(k) = [s_1(k) \ \cdots \ s_N(k)]^\top \in \{-1, +1\}^N. \quad (2)$$

Couplings are encoded by a matrix $J = [J_{ij}]$, with $J_{ij} = 0$ whenever $(i, j) \notin E$. We use a discrete-time Glauber-style update with independent noise terms

$$\eta_i(k) \sim \mathcal{N}(h, T^2), \quad i = 1, \dots, N, \quad (3)$$

TABLE I
MODEL TAXONOMY USED THROUGHOUT THE PAPER.

Model	Name	Representation	Main strength	Main limitation
1	Reference Ising	Microscopic configuration	Highest fidelity	Highest cost
2	Mean-field chain	Coarse observable	Very interpretable	Misses local structure
3	Local-neighborhood probability model	Probability-based, coarse spatial	Retains some spatial information	Mixing distorts transient rates
4	Full-state probability operator	Operator-level/full probability law	Rich probabilistic structure	Exponential state growth
5	Affine probability operator	Structured probability/operator hybrid	Interpretable and bounded	Reduced expressiveness

and conditional probability

$$\mathbb{P}(s_i(k+1) = +1 \mid s(k)) = \sigma \left(\sum_{j=1}^N J_{ij} s_j(k) + \eta_i(k) \right), \quad (4)$$

where $\sigma(x) = 1/(1 + e^{-x})$ is the logistic function. In this formulation, h acts as a mean bias and T controls the spread of the stochastic perturbation. All results in this paper are reported through the induced magnetization process and summary statistics computed over repeated trials.

Equation (4) defines the reference dynamics used throughout the paper. It is the most faithful model considered here and serves as the baseline against which the remaining models are evaluated.

V. MODEL TAXONOMY

To make our comparison precise, Table I describes the type of simplification of each model, along with an example strength and shortcoming. Model 1 is the reference process, and models 2–5 are the simplified surrogates.

VI. SIMPLIFIED MODELS

A. Model 1: Reference dynamics

Model 1 is the stochastic spin system defined in (4). It preserves the full microscopic configuration and serves as the baseline for all metrics, overlays, and parameter translations.

B. Model 2: Mean-field magnetization chain

Model 2 compresses the system to a one-dimensional state based on global magnetization. Let m_k denote the mean spin. Using a scalar effective coupling $\bar{J}$, we evolve

$$m_{k+1} \approx \tanh(\bar{J}m_k + \xi_k), \quad \xi_k \sim \mathcal{N}(h, T^2). \quad (5)$$

This model is cheap, interpretable, and closely connected to mean-field intuition. Its main limitation is that it cannot represent local correlation structure or heterogeneous transients. It preserves a global summary and discards spatial arrangement.

C. Model 3: Local-neighborhood probability model with mixing

Model 3 tracks per-site probabilities while using only local neighborhood information. Let

$$q_i(k) := \mathbb{P}(s_i(k) = +1), \quad i = 1, \dots, N, \quad (6)$$

and define the centered local expectation $2q_j(k) - 1$. A generic neighborhood-based update takes the form

$$\tilde{q}_i(k+1) = \sigma \left(\sum_{j \in \mathcal{N}(i)} \tilde{J}_{ij} (2q_j(k) - 1) + h + \xi_i(k) \right), \quad (7)$$

followed by a mixing step

$$q_i(k+1) = (1 - \alpha)q_i(k) + \alpha \tilde{q}_i(k+1), \quad (8)$$

for some mixing parameter $\alpha \in (0, 1]$. This model preserves a coarse notion of local spatial structure while discarding full configuration-level dependence. The mixing parameter improves numerical behavior and calibration flexibility, but it can alter transient timing.

D. Model 4: Full exponential-state probability operator

Model 4 lifts the system to the full probability simplex over all configurations. Enumerate the 2^N configurations as $\{\omega^1, \dots, \omega^{2^N}\}$ and let

$$x_k \in \Delta^{2^N} \quad (9)$$

be the probability distribution over those configurations. The evolution is linear at the level of the full probability law,

$$x_{k+1} = P x_k, \quad (10)$$

where the matrix $P \in \mathbb{R}^{2^N \times 2^N}$ is constructed from the spinwise conditional update probabilities induced by (4). Equivalently, for each site we may write the conditional heat-bath style probability

$$\mathbb{P}(s_i(k+1) = +1 \mid s(k)) = \sigma \left(\frac{2}{T} \left(\sum_{j \in \mathcal{N}(i)} J_{ij} s_j(k) + h \right) \right), \quad (11)$$

and assemble the resulting full-state transition operator. This model preserves the richest probabilistic structure among the surrogates with the benefit of being linear in the state probabilities, but its state dimension grows exponentially with N and quickly becomes intractable.

E. Model 5: Restricted-interval affine probability operator

Model 5 acts on stacked per-site probabilities under explicit normalization and interval constraints. For each site i , define

$$p_i^+(k) = \mathbb{P}(s_i(k) = +1), \quad p_i^-(k) = 1 - p_i^+(k), \quad (12)$$

and stack these into

$$p_k = [p_1^+(k) \quad p_1^-(k) \quad \dots \quad p_N^+(k) \quad p_N^-(k)]^\top \in [0, 1]^{2N}. \quad (13)$$

The model evolves via a structured affine operator

$$p_{k+1} = A(J, h, T) p_k, \quad (14)$$

where $A(J, h, T)$ is partitioned into 2×2 blocks, following the decomposition

$$A(J, h, T) := \frac{T}{2} \mathbf{1} + (1-T) \left(|h| H(h) + (1-|h|) J^*(J) \right). \quad (15)$$

Here $\mathbf{1}$ is the block-diagonal homogenization operator, $H(h)$ is a block-diagonal field-bias operator, and $J^*(J)$ is a normalized block coupling operator that preserves row stochasticity at the level of each site. Positive couplings favor alignment, negative couplings favor anti-alignment, and larger T pushes the model toward homogenized uncertainty. This model preserves per-site probability structure and interpretability, and is has linear dynamics with state size proportional to the number of spins, but it discards higher-order dependence.

Model 5 is classified as a *hybrid* of the probability-based and operator-level viewpoints. As far as we are aware, it is novel to this topic, and was constructed to be a tractable-size, linear model capable of retaining some local structure.

VII. EXPERIMENTAL PROTOCOL

All models are evaluated under shared initial conditions, common parameter settings, and matched trajectory horizons wherever tractable. The primary observable is magnetization, and repeated trials are used to estimate means and confidence intervals.

The present study uses a 2×3 lattice so that Model 4 remains computationally feasible. For each parameter setting (J, h, T) , we simulate 10 uniquely-initialized trajectories of length 10 timestep for 27 different parameter settings. Unless otherwise stated, initial conditions are shared across models by using the same seeded spin configuration or its induced one-hot probability representation. The representative parameter settings visualized in Fig. 1 include both positive and negative couplings, positive and negative external field, and low- and high-temperature regimes.

For each model and parameter setting, we compute the sample mean trajectory

$$\hat{\mu}_k = \frac{1}{M} \sum_{r=1}^M m_k^{(r)}, \quad (16)$$

and the pointwise 95% confidence interval using the empirical standard error. The full experiment may include a larger parameter grid $\mathcal{P}$, but the figures below highlight representative slices through that grid. This motivates our use of the term “trajectory manifold”, as each trajectory is also a function of the parameters, and not time alone.

A. Cross-model calibration

Different surrogates do not interpret (J, h, T) identically. To compare them on approximately equal footing, we use a two-stage calibration procedure. First, we match source- and target-model summaries by nearest-neighbor search in

behavior space. Second, we fit an affine map between the corresponding parameter sets.

In addition to that automated fit, a small number of global manual calibrations are used as explicit changes of units. Specifically, the temperature parameter of Models 2 and 3 is scaled by a factor of 1.5, the external field of Model 5 is scaled by a factor of 8, and the number of internal updates per reported time step is set to 6 for Model 3 and 8 for Model 4. These calibrations are not hidden tuning; they are part of the comparison and are reported because mismatched parameter semantics are themselves informative.

VIII. COMPARISON METRICS

A. Trajectory RMSE

For a surrogate model q and reference model r , define

$$\text{RMSE}_{\text{traj}}(q, r) := \left(\frac{1}{K+1} \sum_{k=0}^K (\hat{\mu}_k^{(q)} - \hat{\mu}_k^{(r)})^2 \right)^{1/2}. \quad (17)$$

B. Steady-state bias

Using a terminal averaging window $\mathcal{K}_\infty$, define

$$\text{Bias}_\infty(q, r) := \left| \frac{1}{|\mathcal{K}_\infty|} \sum_{k \in \mathcal{K}_\infty} \hat{\mu}_k^{(q)} - \frac{1}{|\mathcal{K}_\infty|} \sum_{k \in \mathcal{K}_\infty} \hat{\mu}_k^{(r)} \right|. \quad (18)$$

C. Variance mismatch

We measure disagreement in variability by

$$\text{VarErr}(q, r) := \frac{1}{K+1} \sum_{k=0}^K \left| \widehat{\text{Var}}(m_k^{(q)}) - \widehat{\text{Var}}(m_k^{(r)}) \right|. \quad (19)$$

D. Convergence-time difference

Let $m_\infty^{(\ell)}$ denote the empirical steady-state mean for model ℓ . For a tolerance $\varepsilon > 0$, define

$$\tau_\varepsilon^{(\ell)} := \min \left\{ k : \left| \hat{\mu}_j^{(\ell)} - m_\infty^{(\ell)} \right| \leq \varepsilon, \forall j \geq k \right\}. \quad (20)$$

The convergence-time difference is then $|\tau_\varepsilon^{(q)} - \tau_\varepsilon^{(r)}|$.

E. Computational cost

Finally, we compare tractability using wall-clock runtime and effective state dimension. The exact numerical values will depend on implementation and hardware. See Table II for runtimes, and the model descriptions in VI for state space sizes.

IX. RESULTS

A. Overview figure

Figure 1 provides our core qualitative trajectory analysis. Each row fixes a parameter setting (J, h, T) and each column corresponds to one model. The plotted quantity is the mean magnetization trajectory with a pointwise 95% confidence interval.

The main qualitative conclusions are visible in Fig. 1. All surrogates reproduce the coarse sign and general scale of the steady-state magnetization in the representative cases shown. However, they differ in three systematic ways. First, Model 5

tends to track the reference transient shape more closely than the other surrogates in the displayed cases. Second, Models 3 and 4 often approach equilibrium more gradually, even when their final magnetization is similar to that of Model 1. Third, Model 4 displays visibly wider confidence bands in several regimes, indicating a larger variance mismatch.

B. Quantitative comparison to the reference model

Table II is designed to complement Fig. 1 with explicit numerical comparisons. The actual numerical values of the modeling error can be improved using a change-of-basis on the parameters, as described in VII-A.

The intended interpretation of Table II is as follows. Trajectory RMSE measures overall transient agreement. Steady-state bias measures long-time agreement. Variance error measures whether the model reproduces uncertainty at the correct scale. Convergence-time difference directly tests the earlier qualitative claim that Models 3 and 4 converge more slowly. Runtime and memory quantify the tractability tradeoff that motivated the comparison in the first place.

C. One-at-a-time parameter sweeps

Figure 1 is useful as a global overview, but it does not isolate which parameter is responsible for a given discrepancy. Figure 2 illustrates what happens to the steady-state magnetization when only one parameter is varied. This directly addresses the concern that a scattered grid of parameter combinations makes model divergence harder to interpret.

These sweeps make it easier to identify which discrepancies are structural and which are merely artifacts of plotting many conditions at once. In particular, they allow one to separate the effect of temperature on variance from the effect of coupling on equilibrium sign and magnitude.

D. Computational tractability

The most faithful surrogate is not automatically the best model for a given task. Model 4 is the richest reduced probabilistic representation in the paper, but its 2^N -state probability vector scales poorly with system size. Model 2 is by far the cheapest but is also the coarsest, with only one internal state. Models 3 and 5 occupy an intermediate position: they preserve more information than the mean-field chain while avoiding full combinatorial explosion.

X. DISCUSSION

The main result of the paper is not that one surrogate dominates the others in every sense. The main result is that representation choice strongly affects *which* aspects of the reference dynamics are preserved.

First, long-time agreement is easier to obtain than transient agreement. Several models reproduce the sign and rough magnitude of the final magnetization while still differing noticeably in convergence rate and transient shape. This matters because a model that looks correct at the last time step can still be misleading if the task depends on transient response.

Second, variance is not a side issue. The wider confidence bands of Model 4 in several parameter regimes show that

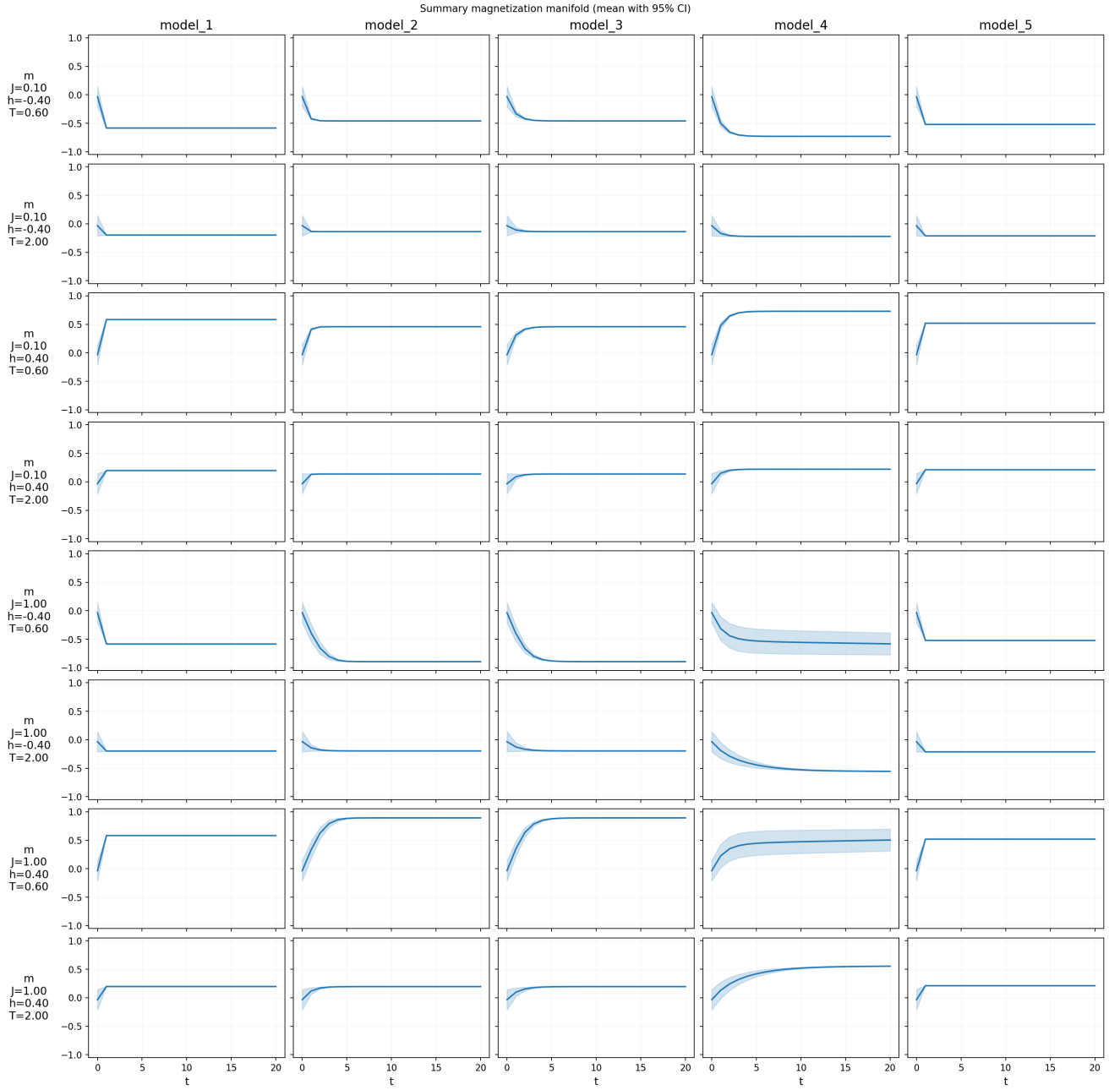


Fig. 1. Summary manifold panels for the five models. Columns correspond to Models 1–5 and rows correspond to representative parameter settings (J, h, T) . Each panel shows the sample mean magnetization trajectory $\hat{\mu}_k$ together with a pointwise 95% confidence interval over repeated trials. The displayed settings include both ferromagnetic and antiferromagnetic couplings, positive and negative external field, and low- and high-temperature regimes, as specified in the left vertical labels .

matching mean behavior does not imply matching uncertainty. Likewise, the smaller state descriptions of Models 2 and 3 can obscure variability structure even when they produce visually plausible average trajectories.

Third, calibration should be treated as part of the scientific result. The need to rescale temperature in Models 2 and 3, field strength in Model 5, and internal update frequency in Models 3 and 4 indicates that these models do not share a common physical meaning for the same nominal parameter values. Rather than hiding those adjustments, we report them

because they clarify what kind of comparison is being made.

Several limitations remain. The comparison is restricted to a small lattice because Model 4 becomes intractable quickly. The main observable is magnetization, which is intentionally coarse and does not fully probe local correlation or domain formation. The representative visualizations emphasize a subset of the full parameter grid. Finally, the present paper does not claim rigorous stability results in the control-theoretic sense; instead, it compares transient and equilibrium behavior empirically.

TABLE II
QUANTITATIVE COMPARISON OF MODELS 2–5 AGAINST THE REFERENCE MODEL 1. ALL VALUES ARE AVERAGES ACROSS ALL PARAMETERIZATIONS AND INITIAL CONDITIONS FOR EACH MODEL.

Model	RMSE _{traj}	Bias _∞	VarErr	$ \Delta\tau_\varepsilon $	Runtime
Model 2: Mean-field chain	0.062112	-2.848635e-03	0.007080	0.666667	0.000010s
Model 3: Neighborhood probability + mixing	0.066929	-1.972651e-03	0.007801	0.962963	0.001140s
Model 4: Full-state probability operator	0.108910	-6.876069e-03	0.041820	3.444444	0.178714s
Model 5: Affine probability operator	0.027885	-8.659184e-11	0.000133	0.000000	0.000352s

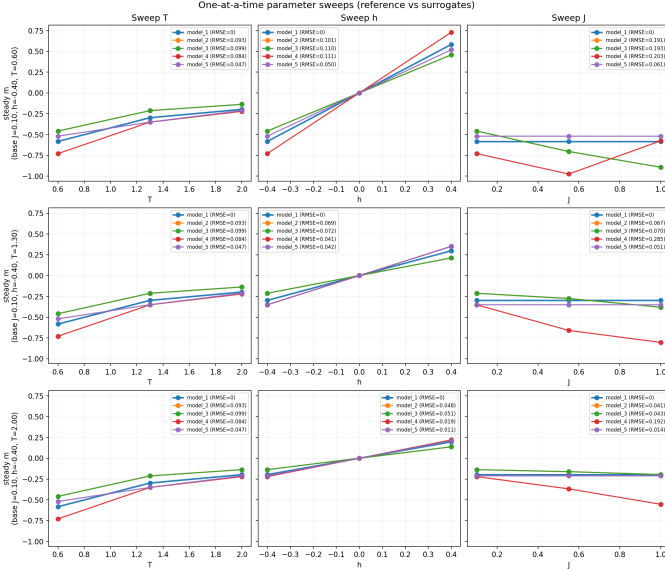


Fig. 2. One-at-a-time parameter sweeps. Example layout: top row sweeps temperature T for fixed (J, h) , middle row sweeps external field h for fixed (J, T) , and bottom row sweeps coupling J for fixed (h, T) . Each subplot overlays the reference model with the four surrogates and may report trajectory RMSE in the legend.

TABLE III
INTERPRETIVE SUMMARY OF THE QUALITATIVE TRACTABILITY-FIDELITY TRADEOFF. COST SUMMARY IS THE RUNTIME AS A MULTIPLE OF MODEL 1, FOR THE 3-BY-3 ATOM LATTICE USED IN THESE TESTS.

Model	State dimension	Cost summary (time÷Model 1)	Practical interpretation
1	2^N implicit MC simulation	1.0 (3.4e-5s)	Reference fidelity, no compact state
2	1	0.36x	Cheapest global summary
3	N site probabilities	33.3x	Retains local structure at moderate cost
4	2^N explicit distribution	5215.8x	Richest surrogate, least scalable
5	$2N$ stacked probabilities	10.3x	Structured middle ground

XI. CONCLUSION

This paper compared five representations of the same discrete Ising dynamics within a single experimental framework: one reference model and four surrogates spanning coarse observable, probability-based, and operator-level viewpoints.

The empirical takeaway is that agreement in final magnetization does not imply agreement in transient rate, variance, or computational cost, even when modeling the “same” system. Model 2 is the most interpretable and cheapest, but it discards local structure. Model 4 is the richest probabilistic surrogate, but scales poorly. Models 3 and 5 occupy intermediate positions, with Model 5 providing a particularly interpretable structured compromise in the representative cases shown here. The appropriate model therefore depends on what one needs to preserve: coarse trend, transient shape, uncertainty, or tractability.

Future work should expand the parameter grid, include biased and clustered initial conditions, evaluate additional observables beyond magnetization, and replace the present global calibrations with a more systematic identification procedure. The broader lesson is that simplification is not a single algorithm but a modeling choice, and that the consequences of that choice become clearest only under close and critical examination.

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